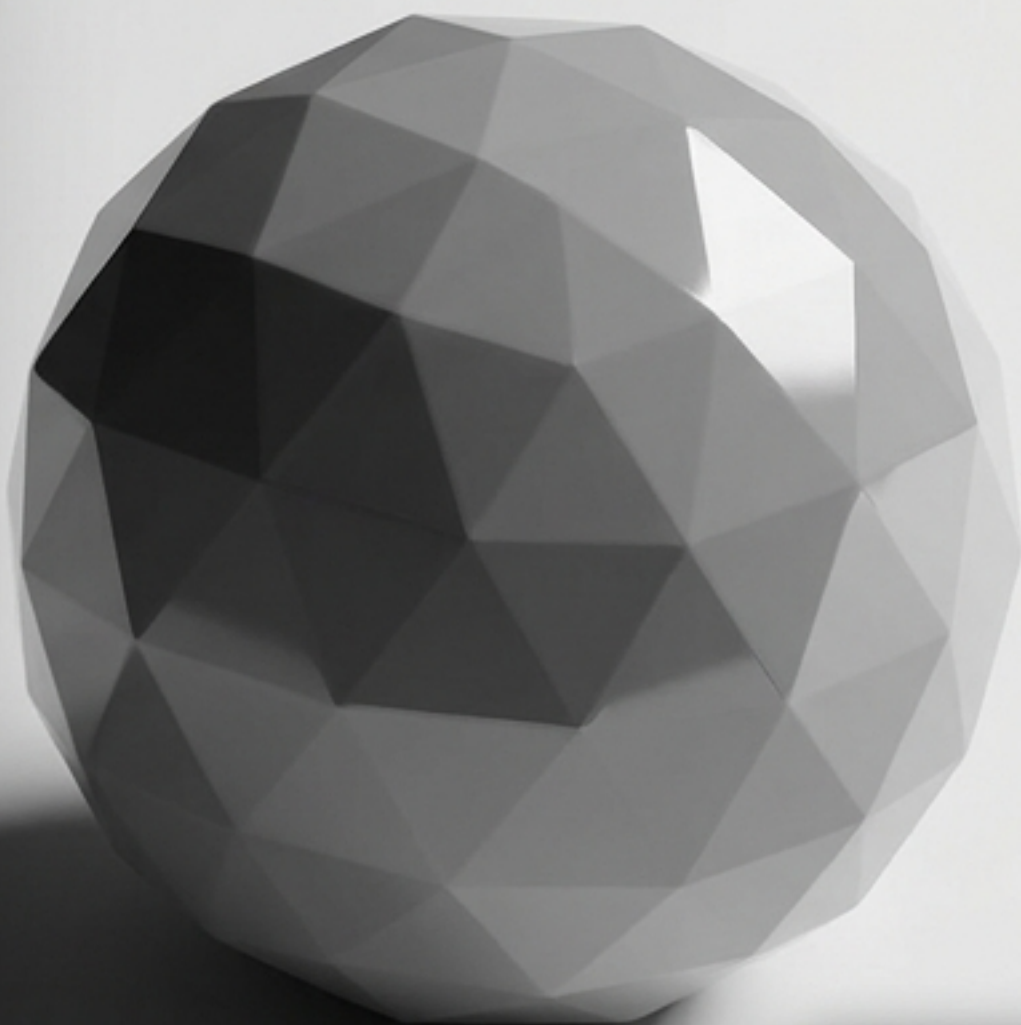


LOOK-DEV
LABORATORY

REFLECTANCE:
0.95

ROUGHNESS:
0.02



ROUGHNESS:
0.02

POLYGON COUNT:
4096

LIGHT SOURCE:
DIRECTIONAL 90°

The ClayBound Surface Grammar

A step-by-step technical isolation of Macro, Meso, and Micro handcrafted details in Blender 4.x.

CORE DIRECTIVE: Translating 'soft,' 'handmade,' and 'cozy' into measurable, reproducible technical parameters.



ANALYSIS COMPLETE: 100%

SURFACE:
CLAY_V4

DISPLACEMENT:
HAND-SCULPT

ROUGHNESS:
0.85



ROUGHNESS:
0.85

LIGHT SOURCE:
STUDIO SOFTBOX

MATERIAL:
DIGITAL_CLAY_MATTE

THE SCULPTOR'S WORKBENCH

TARGET VISUAL IDENTITY



1



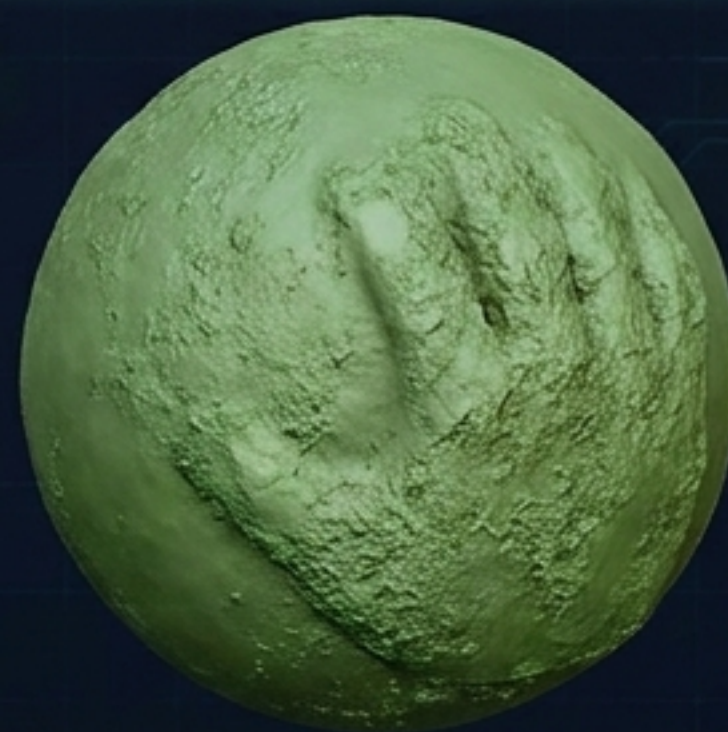
MACRO (Geometry)

2



MESO (Deformation)

3



MICRO (Grain)

THE GOLDEN RULE: Handmade does not mean dirty. Never rely on generic noise overlays, photoreal scratches, or fake repetitive fingerprint stamps.

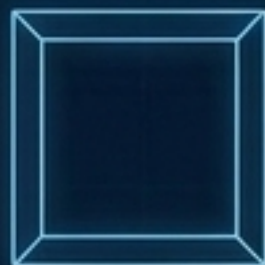
METHODOLOGY: Standardizing the visual subject. By utilizing a single test sphere, we eliminate character-geometry cognitive load and isolate exact shader mechanics.

STANDARD CGI

DIGITAL CLAY

EDGES

Mathematical
90° angles.



Soft, rounded chamfers;
pliable silhouettes.



SURFACE

Uniformly smooth
planes.



Organic massing,
broad low-angle tool
marks and thumb presses.



REFLECTIONS

Clearcoat, glossy, wet.



Velvet-matte, dry,
strict limits on specular
highlights.



DETAIL GENERATION

High-frequency procedural noise.

Low-amplitude procedural layers
baked to tangent-space normal maps.



MACRO SCALE: Stripping the Mathematical Edge

OBJECTIVE: Silhouetted massing and primary volume softening.

PREREQUISITE: Apply all transforms (Scale must equal 1.0, 1.0, 1.0).

▼  Bevel Modifier     ▲ × ⋮

Width: 0.005m | Segments: 2 | Limit Method: Angle (30°)

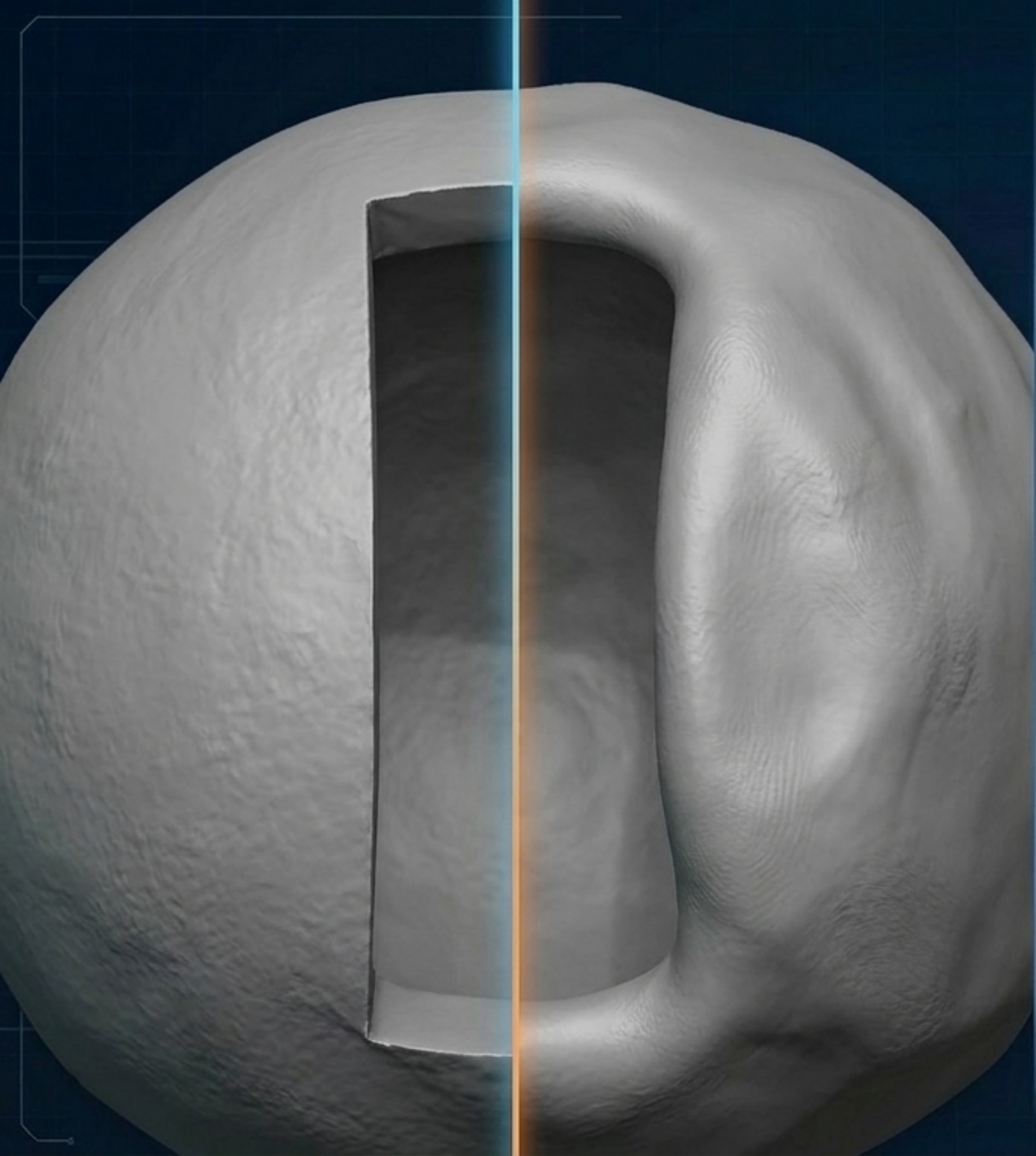
▼  Subdivision Surface     ▲ × ⋮

Render Levels: 2 | Viewport Levels: 1

▼  Weighted Normals     ▲ × ⋮

Keep Sharp: Enabled

RULE: Pliable volume must exist in the physical mesh before any light transport or material logic is applied.

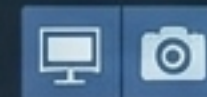


MACRO/MESO BOUNDARY: Faking the Join

THE CHALLENGE: Rounding low-poly interior seams to fake the look of clay pushed together by a thumb, without generating complex supporting geometry.

IMPACT: Procedurally perturbs normals along sharp edges, saving VRAM while maintaining the illusion of hand-kneaded seams.

▼ IMPLEMENTATION



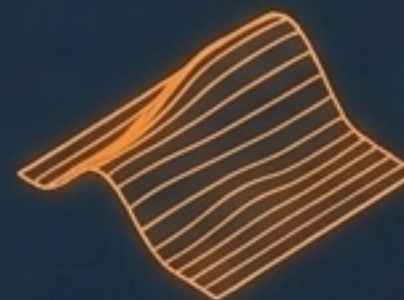
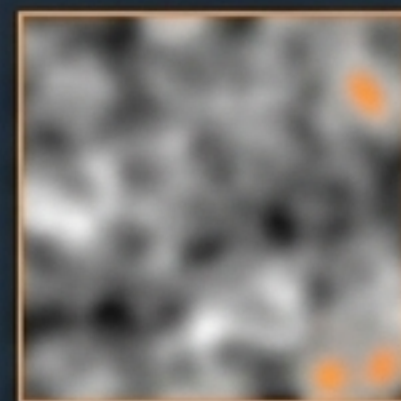
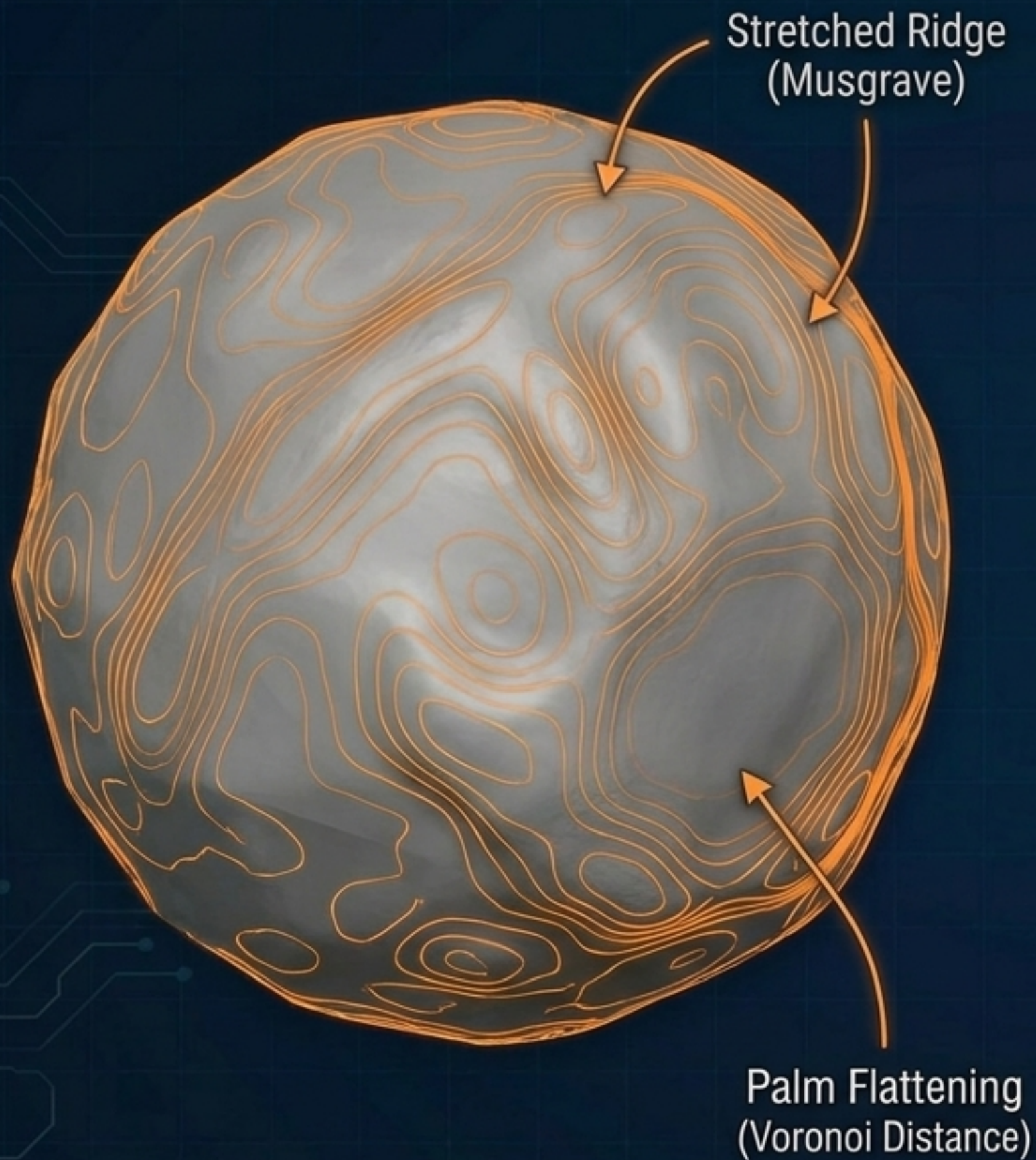
Node: Bevel Shader 

Radius: 0.010m

Samples: 8

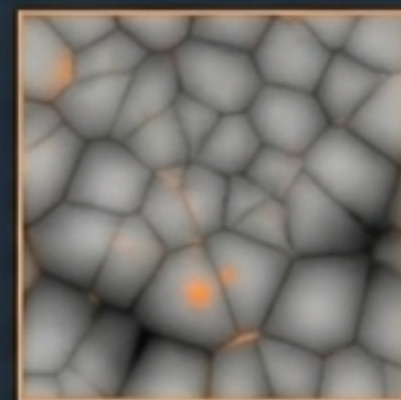
MESO SCALE: Broad Form Disruption

OBJECTIVE: Introduce faint contour variations that establish scale without overwhelming the primary asset silhouette.



1. The Folds

Simulated via Musgrave/Noise textures. Scaled anisotropically and mapped to Bump.



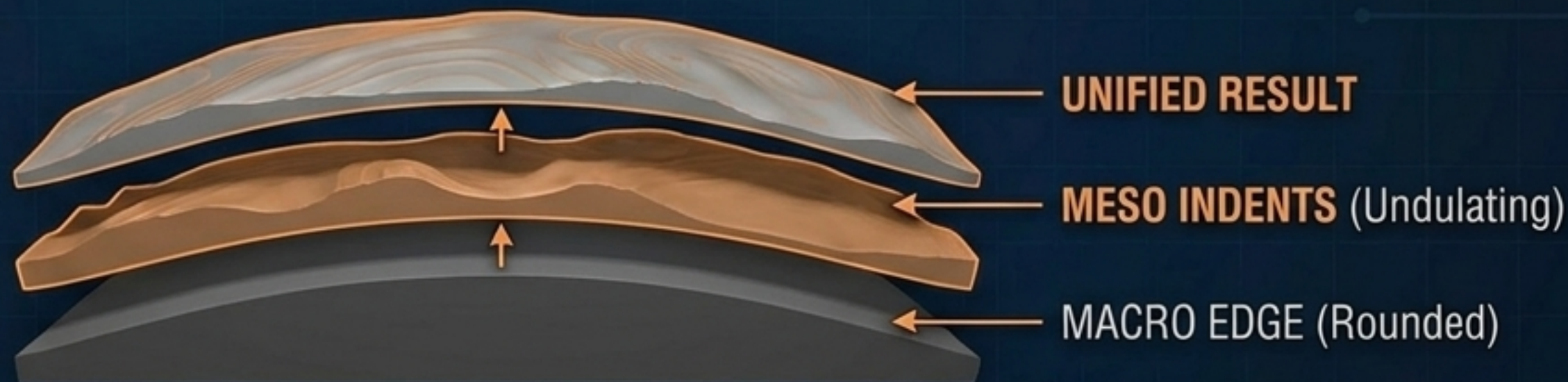
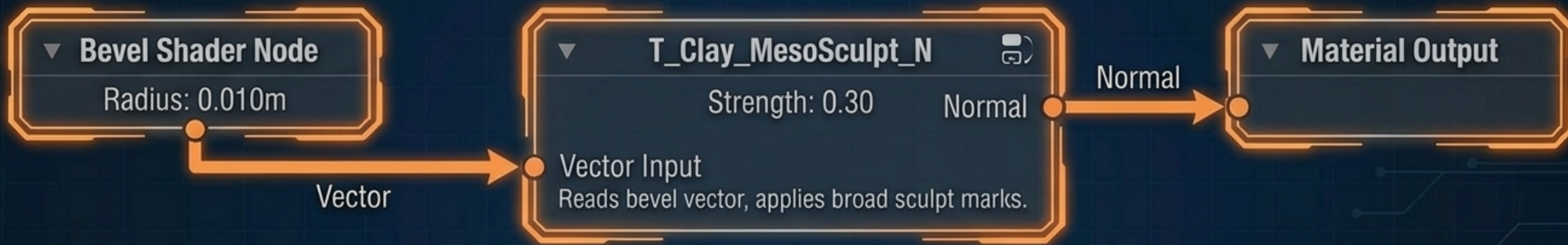
2. The Flattenings

Simulated via Voronoi (Distance) texture. Mimics the flat press of a palm or thumb.

RULE: Keep variations subtle. Emulate broad tools and low-angle strokes.

MESO SCALE: Vector Integration

THE DANGER: Naive normal mapping overwrites geometric bevels, destroying the edge-softening work done in the Macro phase.



NOTE: The Meso Normal map is baked from a high-poly sculpt pass utilizing wide, flat brush strokes (Clay Strips/Flatten).



MICRO SCALE: Porosity & Starch Grain

- **OBJECTIVE:** Replicate the uniform porosity of unglazed modeling clay at the millimeter scale.
- **IMPLEMENTATION:** T_Clay_MicroGrain_N. Generated via dense procedural Voronoi/Perlin passes, baked into a normal map.
- **PARAMETER:** Normal Map Strength 0.15.

THE ABSOLUTE RULE

World-Space / Object-Space



Causes texture 'swimming' when skeletal mesh deforms during animation.

Tangent-Space UV Mapping



Strictly required. Locks micro-grain stability to the physical surface.

MICRO SCALE: Reflectivity Rules

THE PHILOSOPHY: Clay reflects light through micro-scattering, not direct specular bouncing.



PRINCIPLED BSDF PARAMETER LOCKS

Parameter	Value / Range	Description
Roughness	0.65 to 0.85	Modulated using <code>T_Clay_MesoRoughness_R</code> to allow broad highlights along sculpt lines.
Specular IOR Level	Locked at 0.450	Tames direct reflections and maintains a 'dry' look.
Coat Weight	Strictly 0.000	Zero clearcoat disables the glossy ceramic glaze look entirely.
Metallic	Strictly 0.000	Strictly dielectric material.

MATERIAL RESPONSE: Red-Shifted Subsurface Scattering



WHY IT MATTERS:

Digital clay is not solid stone. To feel 'alive,' it must absorb and scatter light internally, especially at thin extremities (ears, fingers, sharp edges).

THE SETUP

Method: Random Walk

Delivers accurate light transport across high-curvature features

Subsurface Weight: 0.250

Subsurface Scale: 0.020m

Controls scatter depth relative to character size

Subsurface Radius:

R: 1.000

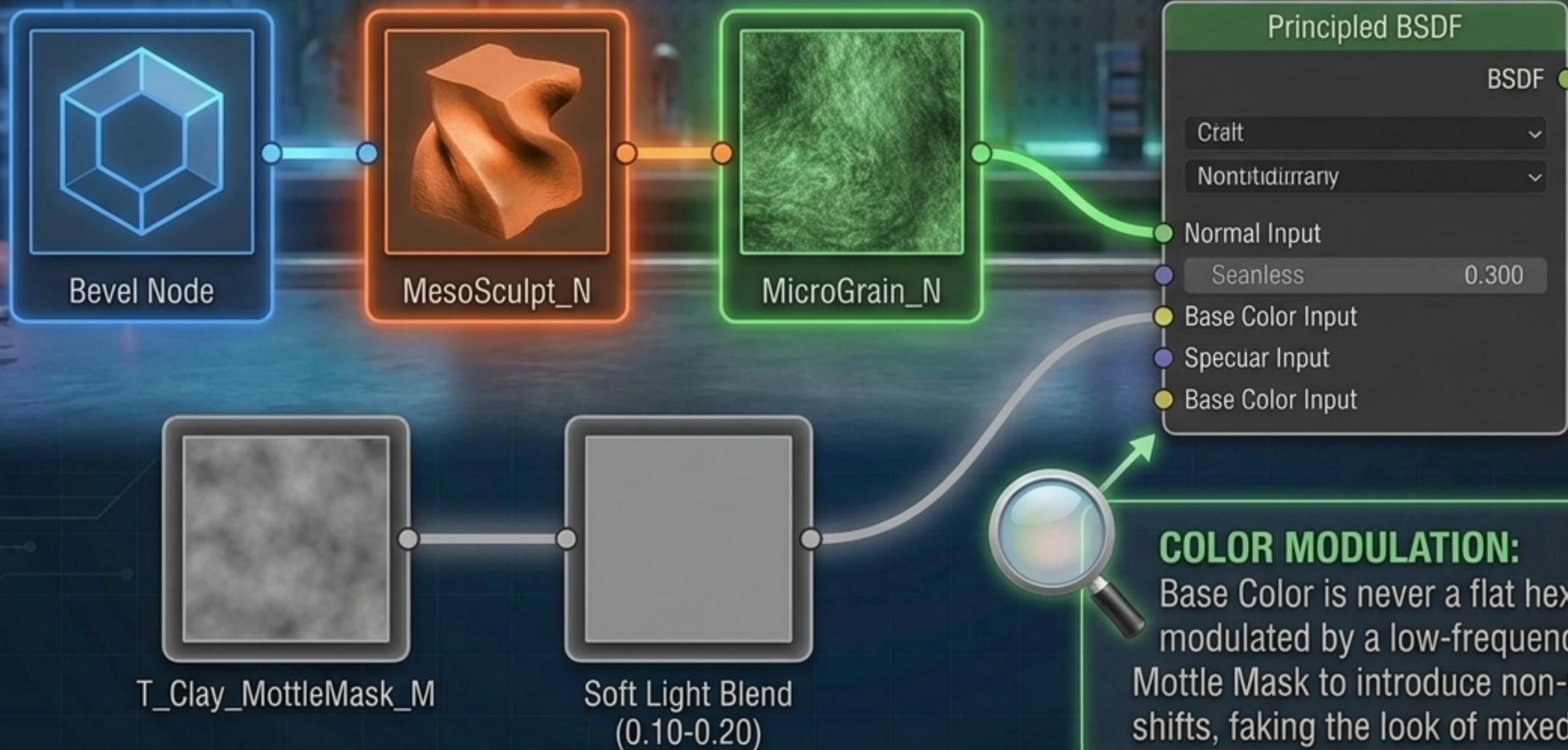
G: 0.350

B: 0.200

RESULT: This specific red-shifted vector creates a warm internal glow by absorbing blue/green wavelengths.

SYNTHESIS: The Master Shader Architecture

The ultimate cheat sheet: balancing render efficiency with tactile realism.



THE SCULPTOR'S WORKBENCH

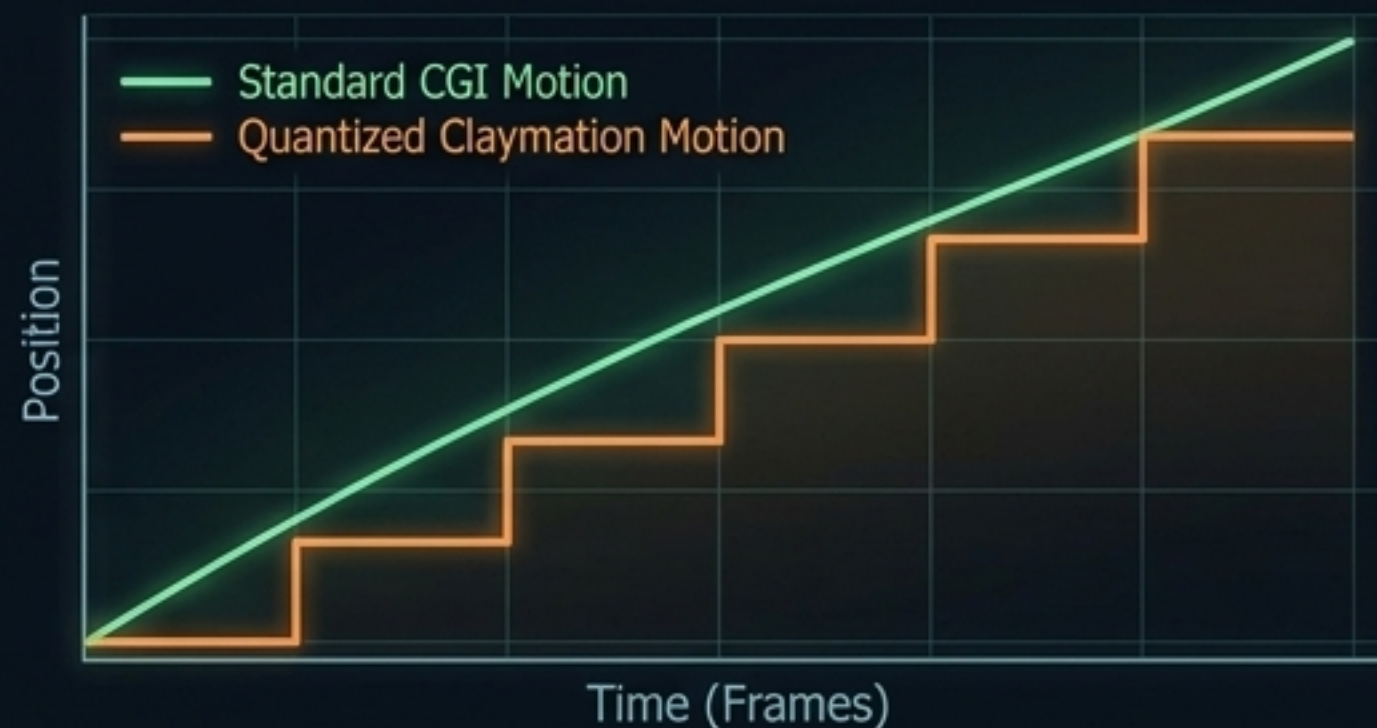
PHYSICS DASHBOARD



ANIMATION DYNAMICS: The Stop-Motion Jitter

PHYSICS DASHBOARD

THE GOAL: Emulate the discrete surface re-sculpting of an animator's thumb presses over time, without fluid sliding.



DRIVER FORMULA:

$$\text{Offset_X} = \lfloor \text{Frame} / N \rfloor \times C$$

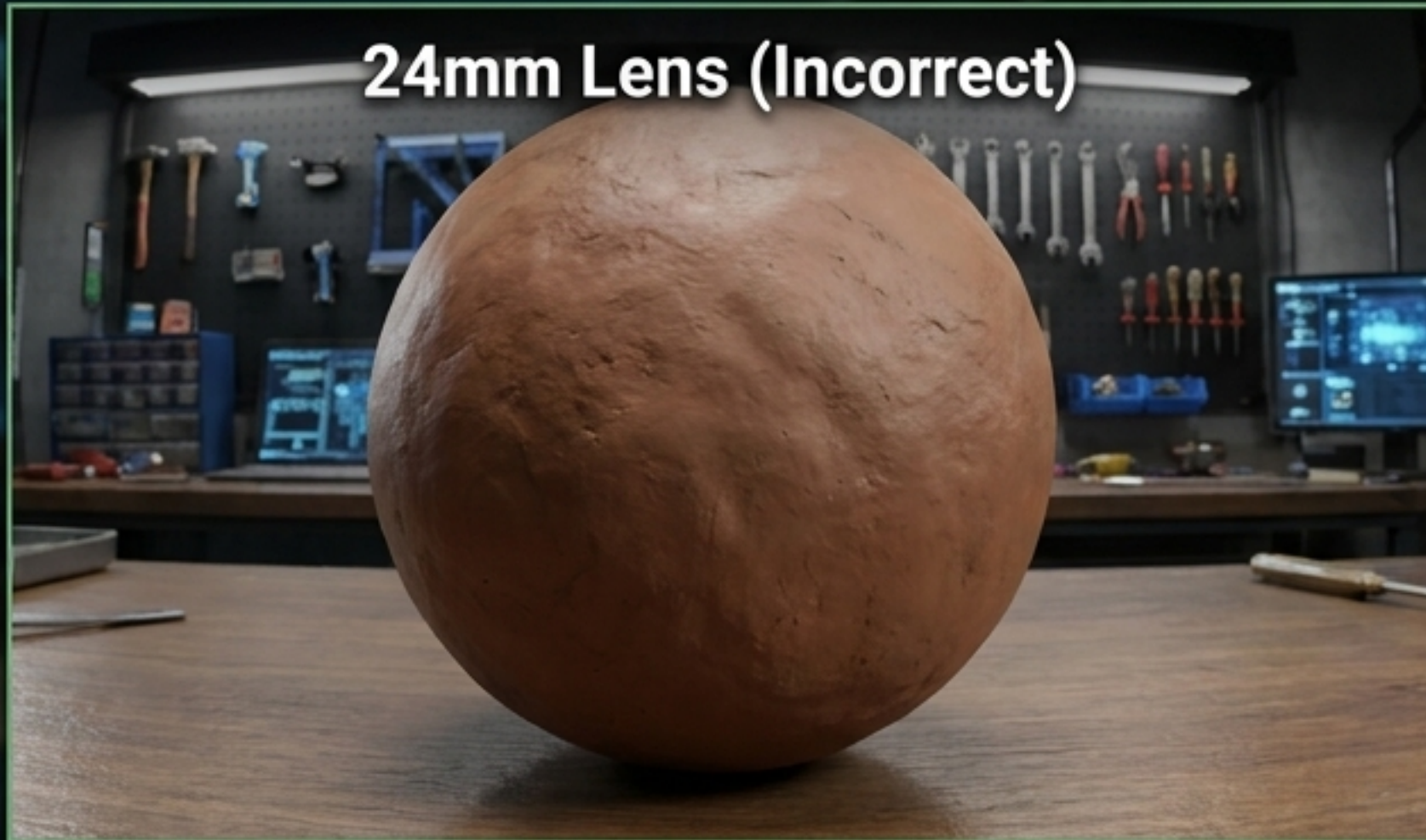
N = Frame step interval (e.g., animating on twos, $N=2$).
 C = Constant spatial step offset.

EXECUTION:

Applied to mapping location offsets. The surface sculpt updates discretely every N frames, but remains locked to the Tangent-space UVs in between, completely preventing texture swimming.

PRESENTATION: Lens Compression & Miniature Scale

THE ILLUSION: To make digital assets feel like physical, hand-sculpted miniatures, the camera must mimic real-world macro photography.



LENS SELECTION:

85mm to 105mm. Long focal lengths compress perspective and reduce wide-angle geometric distortion.

DEPTH OF FIELD:

f/4.0 to f/5.6. Softens distant objects to enforce the perceptual illusion that the subject is only a few inches tall.

PRESENTATION: Studio Illumination & Color



KEY LIGHT: 5200K
(Warm main illumination, 250W)

RIM LIGHT: 7000K
(Cool, sharp accent highlighting the red-shifted subsurface glow, 350W)

FILL LIGHT: 6500K
(Cool ambient shadow fill, 75W)

COLOR MANAGEMENT (Blender 4.x)

View Transform: AgX.
Crucial for preventing harsh saturation clipping in high-intensity subsurface areas.

Look: Medium High Contrast.
Separates soft shadowed recesses from illuminated matte planes.

PRESENTATION: Look-Dev Laboratory



THE VALIDATION CHECKLIST

- [✓] **MACRO:** Object scale applied (1.0, 1.0, 1.0) and Bevel/Subdiv modifiers active?
- [✓] **MESO:** Normal vector chains routing Bevel -> Sculpt -> Micro correctly?
- [✓] **MICRO:** Tangent-space UVs utilized to guarantee zero texture swimming?
- [✓] **MATERIAL:** Dielectric confirmed (Metallic 0.0, Coat 0.0, Roughness > 0.65)?
- [✓] **LIGHT:** AgX Color management active with rim-lighting to trigger SSS?

FINAL PRODUCTION RULE: Let the lighting, the modifiers, and the subtle normal imperfections do the work. Visual trust is broken by noisy clutter. Keep it clean. Keep it tactile.